

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First Semester of B. Tech (All Branches) Examination****Apr-May 2018****EC101 Basics of Electronics Engineering****Date: 14.05.2018, Monday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1.(a)** Explain V-I Characteristics of PN Junction diode. [03]
- (b)** State and explain Ohm's Law. [03]
- Q - 2.(a)** With neat and clean diagram, explain Full-wave rectifier and derive its equation for rectification efficiency. [07]
- Q- 2.(b)** Answer any two questions. [08]
- (i) Derive an expression for the capacitance of a parallel plate capacitor.
 - (ii) Derive the equations of E.M.F for transformer.
 - (iii) Explain Star/Delta transformation of resistors.
- Q - 3** Answer any two questions. [14]
- (a) Obtain 1's and 2's complements of following binary numbers.
1010101, 0111000, 0000001, 100000.
 - (b) Implement following using basic gates.
 - (i) $XY + XY'$
 - (ii) $(X + Y)(X + Y')$
 - (iii) $XYZ + X'Y + XYZ'$
 - (iv) $ABC + A'B'C' + A'BC + ABC' + A'B'C'$
 - (c) Convert Binary 11010111.110 to decimal, octal and hexadecimal.

SECTION – II

- Q - 4.(a)** Explain charging of a capacitor and derive the equation for charging voltage across a capacitor. [07]
- Q – 5. (a)** Answer any two questions. [10]
- (i) Define Electric Field, Electric flux Density and Electric field Intensity and derive the relationship between them.

(ii) Describe the action of the following filter circuits: (1) Capacitor filter (2) π – filter.

(iii) Derive the relationship between α and β with reference to transistor.

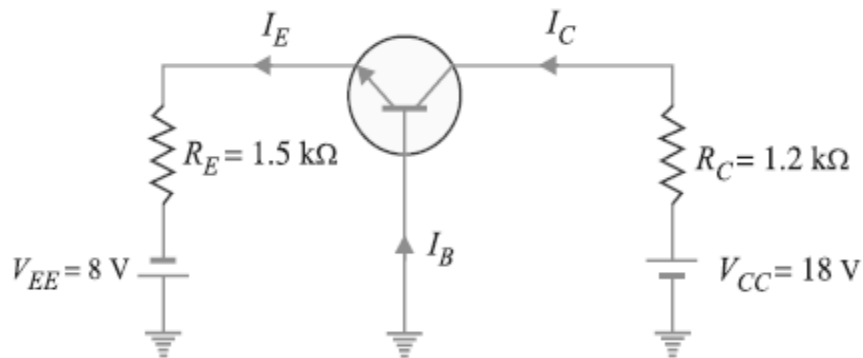
Q – 5.(b) Describe the factors affecting the value of capacitance of a capacitor. [04]

OR

(b) Explain working of Transistor in detail. [04]

Q – 6 Answer any two questions. [14]

(i) For common base circuit shown in Figure. Determine I_C and V_{CB} . Assume transistor to be of silicon.



(ii) Derive an equation for energy stored in a capacitor.

(iii) What is a Zener diode? Draw the equivalent circuit of an ideal zener in breakdown region.

(iv) Draw the input and output characteristics of common Emitter (CE) connection.
